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Platform: OPNsense 26.1.6
Virtualization: VMware Workstation Player 17
Protocol Used: IPsec IKEv2 with Pre-Shared Key (PSK)

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1. Objective

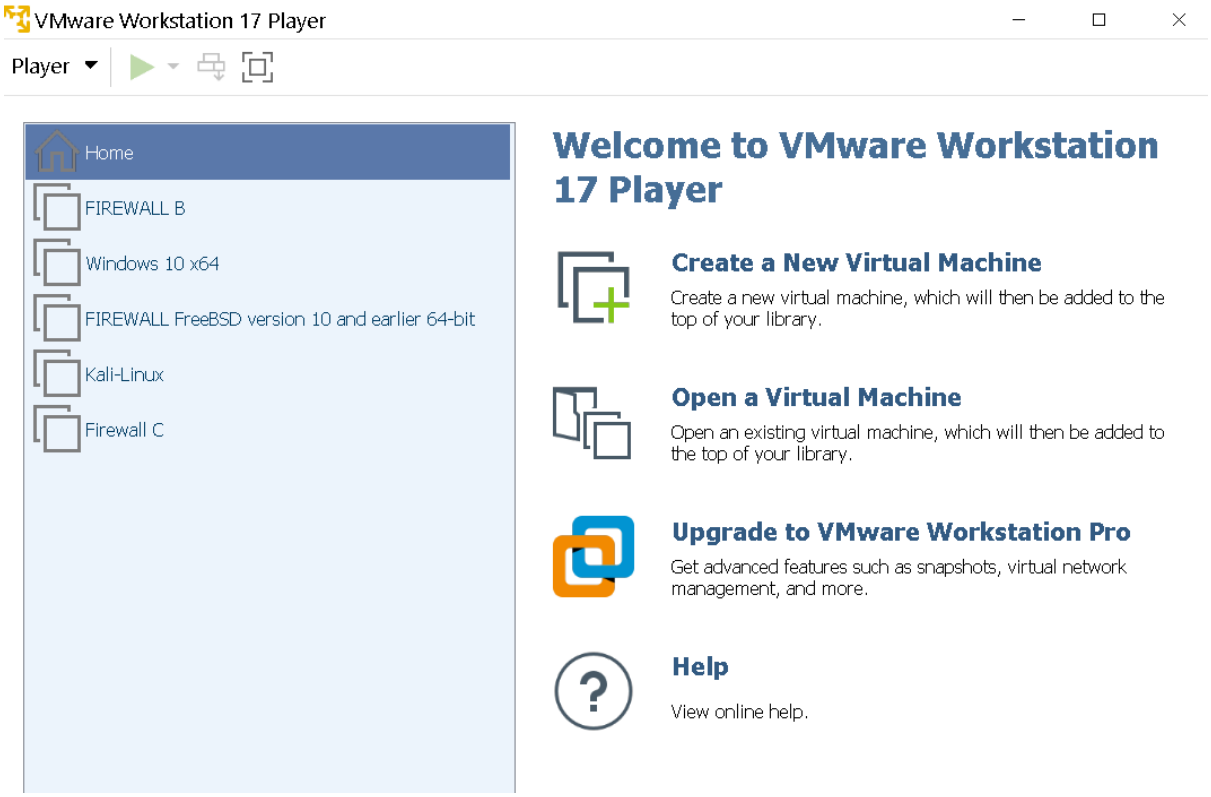
"The objective of this report is to document the step-by-step configuration of a Site-to-Site IPsec VPN tunnel between two OPNsense firewalls (Site A and Site B) using IKEv2 protocol with Pre-Shared Key (PSK) authentication. The tunnel connects two isolated virtual networks simulating a real enterprise environment similar to Tata Electronics where multiple country offices are connected via secure IPsec tunnels."

2. Lab Topology & IP Plan

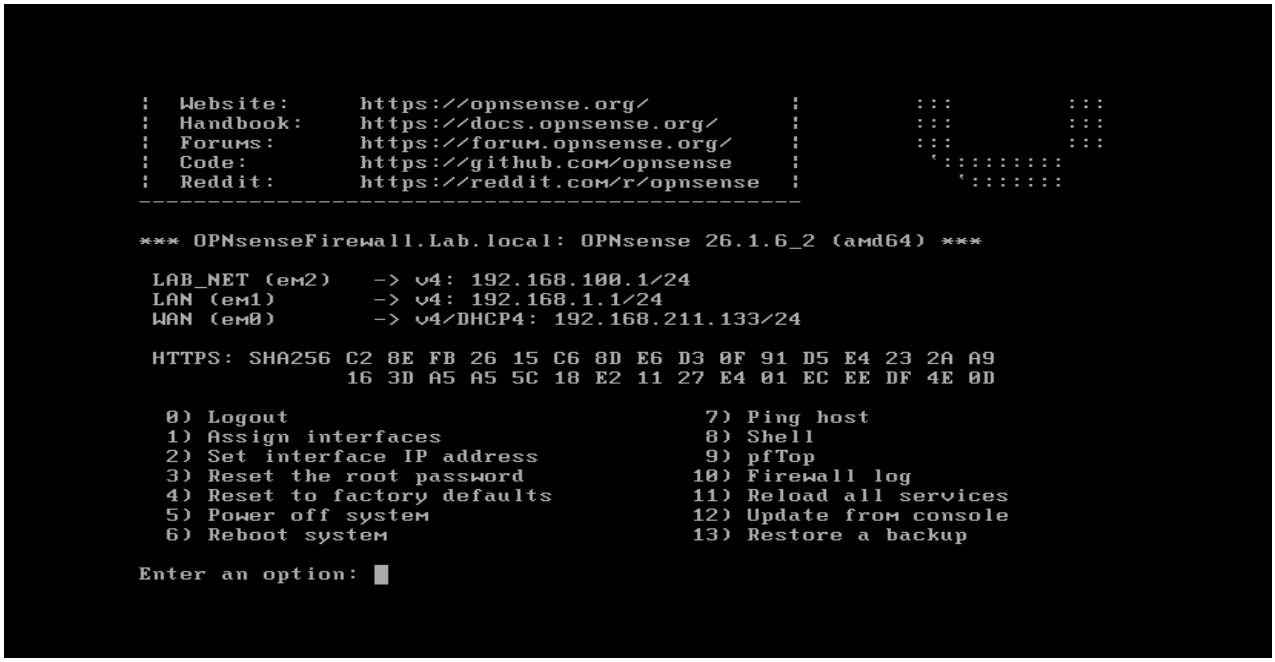
Device	Interface	IP Address	Role
OPNsense A	WAN	192.168.211.133	Site A Firewall
OPNsense A	LAN	192.168.1.1	Site A Office Network
OPNsense A	LAB_NET	192.168.100.1	Site A Lab Network
OPNsense B	WAN	192.168.211.134	Site B Firewall
OPNsense B	LAN	192.168.2.1	Site B Office Network
OPNsense B	LAB_NET	192.168.200.1	Site B Lab Network
Kali VM	LAB_NET	192.168.100.100	Test Machine
Windows VM	LAB_NET	192.168.100.101	Test Machine

Screenshots:

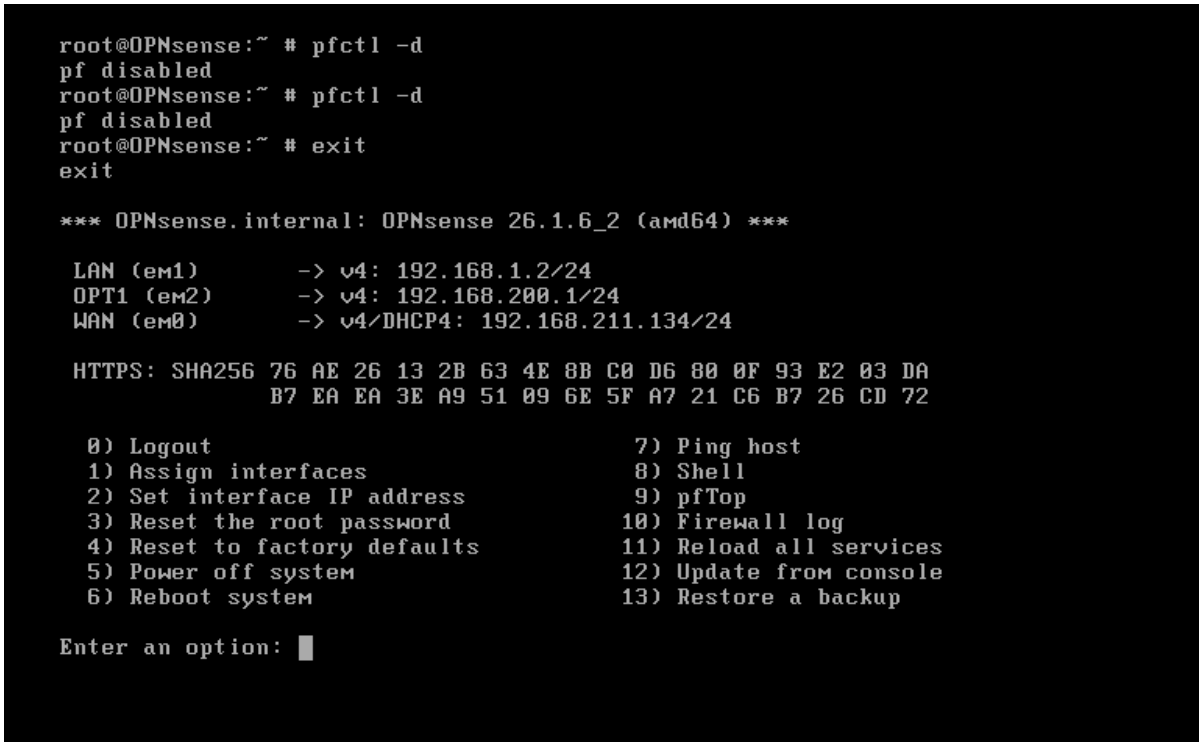
VMware Player showing all VMs:



Site A Firewall Console page:



Site B Firewall Console page:



3. Tools & Software Used

Tool	Version	Purpose
OPNsense	26.1.6	Firewall & VPN
VMware Workstation Player	17 (free)	Virtualization
Kali Linux	Latest	Testing & Verification
Nmap	Built-in Kali	Port Testing
Web Browser	Edge	Firewall GUI Access

4. Pre-requisites

"Before configuring the IPsec tunnel the following were completed:"

- Two OPNsense VMs created in VMware Player 17
- Each VM configured with 3 network adapters (NAT, VMnet1/3, VMnet2/4)
- OPNsense installed on both VMs
- Interfaces assigned (em0=WAN, em1=LAN, em2=OPT1)
- IP addresses configured on all interfaces
- Basic connectivity verified between VMs
- Kali VM able to ping gateway (192.168.100.1)

VM Creation:

"Two virtual machines were created in VMware Player 17 for Site A and Site B firewalls. Each VM was configured with the following hardware settings:"

Setting	**OPNsense A**	**OPNsense B**
RAM	2 GB	1 GB
Hard Disk	20 GB	20 GB
Processors	2	1
Network Adapter 1	NAT	NAT
Network Adapter 2	VMnet1	VMnet3
Network Adapter 3	VMnet2	VMnet4
ISO File	OPNsense 26.1.6	OPNsense 26.1.6

"Note: VMware Player 17 free version does not support Virtual Network Editor GUI. VMnet3 and VMnet4 were automatically created when assigned to OPNsense B adapters."

Interface Assignment:

"After installation, interfaces were assigned on both firewalls using the OPNsense console menu (Option 1 — Assign Interfaces)."

Site A Interface Assignment:

Interface	Assigned To	Role
em0	WAN	Internet facing (NAT)
em1	LAN	Office network (VMnet1)
em2	OPT1/LAB_NET	Lab network (VMnet2)

Site B Interface Assignment:

Interface	Assigned To	Role
em0	WAN	Internet facing (NAT)
em1	LAN	Office network (VMnet3)
em2	OPT1/LAB_NET	Lab network (VMnet4)

IP Address Configuration

"Static IP addresses were configured on LAN and OPT1 interfaces of both firewalls using console Option 2 — Set Interface IP Address. WAN interface receives IP automatically from VMware NAT (DHCP)."

Site A IP Configuration:

Interface	**IP Address**	**Subnet**	**DHCP Range**
WAN (em0)	192.168.211.133	/24	Automatic
LAN (em1)	192.168.1.1	/24	192.168.1.100-200
OPT1 (em2)	192.168.100.1	/24	192.168.100.100-200

Site B IP Configuration:

Interface	IP Address	Subnet	DHCP Range
WAN (em0)	192.168.211.134	/24	Automatic
LAN (em1)	192.168.2.1	/24	192.168.2.100-200
OPT1 (em2)	192.168.200.1	/24	192.168.200.100-200

"Note: Each network uses a unique IP range to avoid conflicts. WAN IPs are used for tunnel endpoints as they represent the internet-facing side of each firewall."

Pre-Shared Key (PSK) Configuration

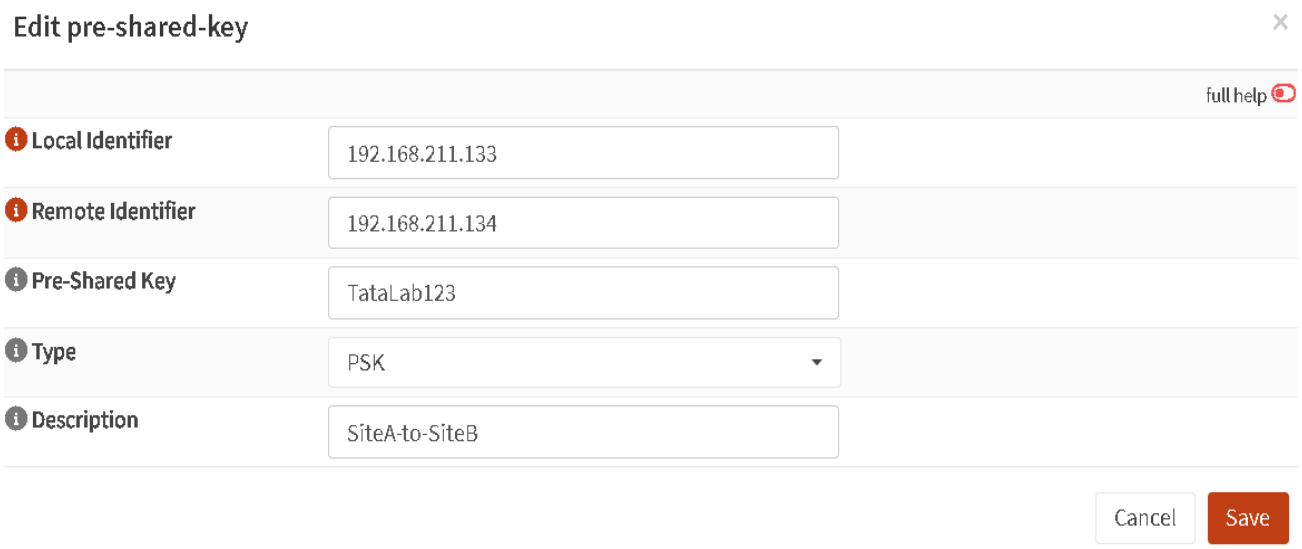
"A Pre-Shared Key was configured on both firewalls. The PSK acts as a password that both firewalls use to authenticate each other during IKE Phase 1 negotiation. Both sides must have identical PSK otherwise the tunnel will be rejected."

What is PSK?

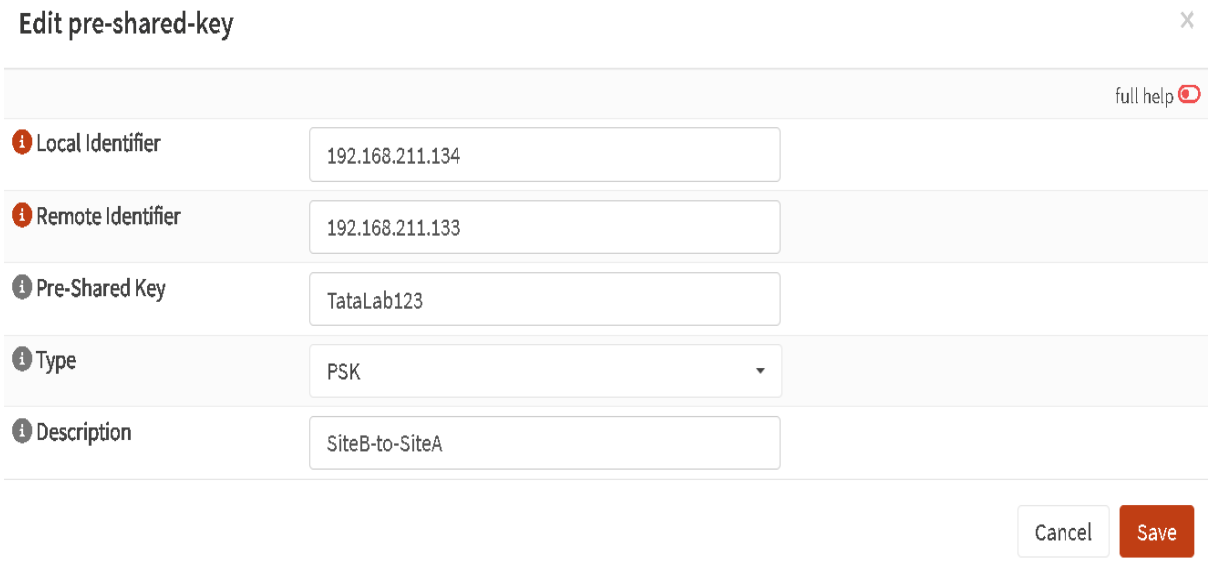
"Pre-Shared Key is a secret key manually configured on both ends of the tunnel. It is used to verify the identity of the remote firewall before establishing the encrypted tunnel."

Configuration on Site A:

- Go to **VPN → IPsec → Pre-Shared Keys → Add**



Configuration on Site B:



IPsec Connection & Authentication Configuration

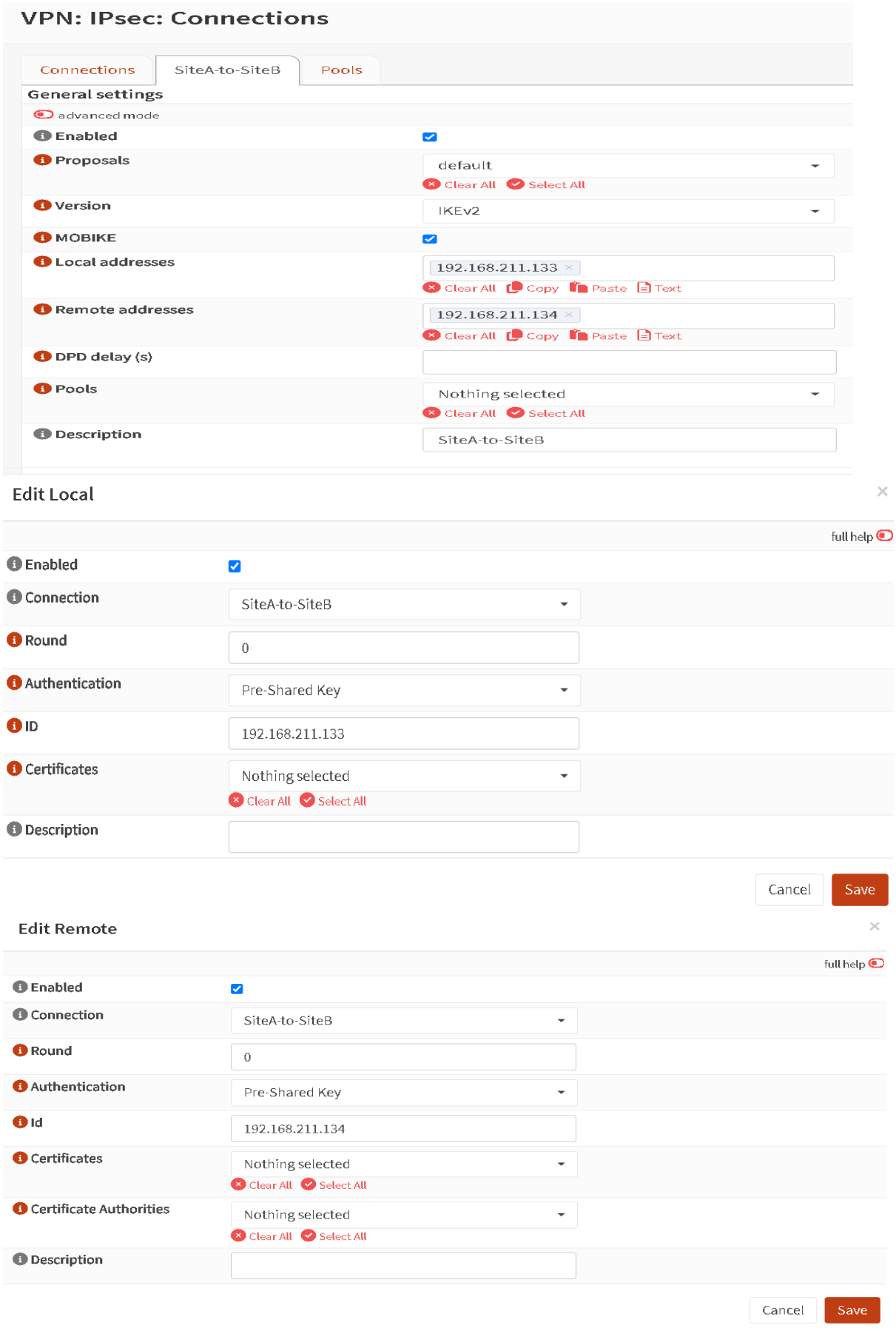
"IPsec connections were created on both firewalls. The connection defines which two firewalls will communicate, which IKE version will be used, and how each firewall will authenticate itself to the other."

What is IKEv2?

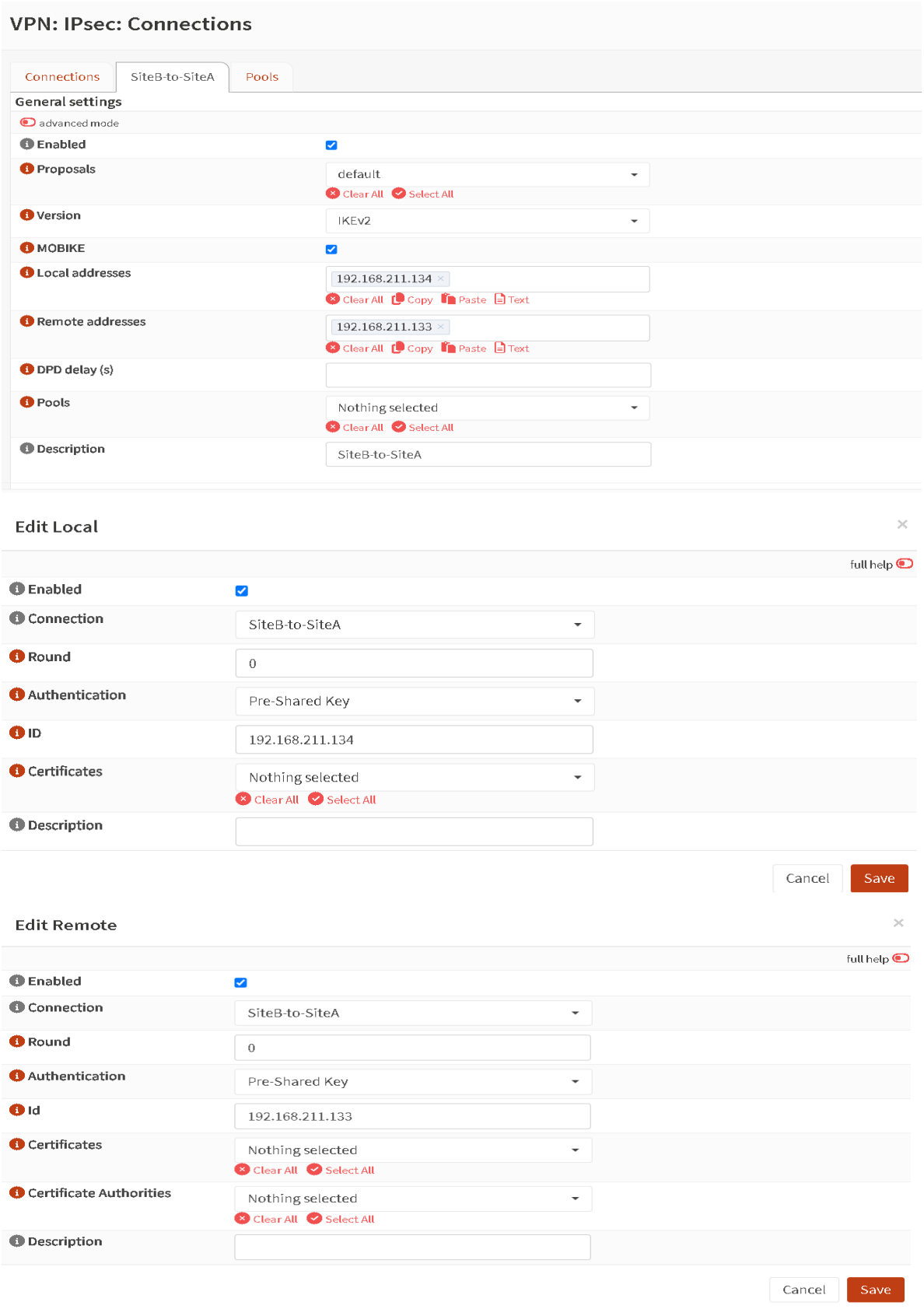
"IKE (Internet Key Exchange) version 2 is the protocol used by both firewalls to negotiate and establish the secure tunnel. It handles authentication, encryption negotiation and key exchange between the two sites."

Configuration on Site A:

Go to **VPN → IPsec → Connections → Add**


VPN: IPsec: Connections
Connections
SiteA-to-SiteB
Pools
General settings
advanced mode
Enabled
Proposals
default
Clear All
Select All
Version
IKEv2
MOBIKE
Local addresses
192.168.211.133
Clear All
Copy
Paste
Text
Remote addresses
192.168.211.134
Clear All
Copy
Paste
Text
DPD delay (s)
Pools
Nothing selected
Clear All
Select All
Description
SiteA-to-SiteB
Edit Local
full help
Enabled
Connection
SiteA-to-SiteB
Round
0
Authentication
Pre-Shared Key
ID
192.168.211.133
Certificates
Nothing selected
Clear All
Select All
Description
Cancel
Save
Edit Remote
full help
Enabled
Connection
SiteA-to-SiteB
Round
0
Authentication
Pre-Shared Key
Id
192.168.211.134
Certificates
Nothing selected
Clear All
Select All
Certificate Authorities
Nothing selected
Clear All
Select All
Description
Cancel
Save

Configuration on Site B:



Child SA Configuration

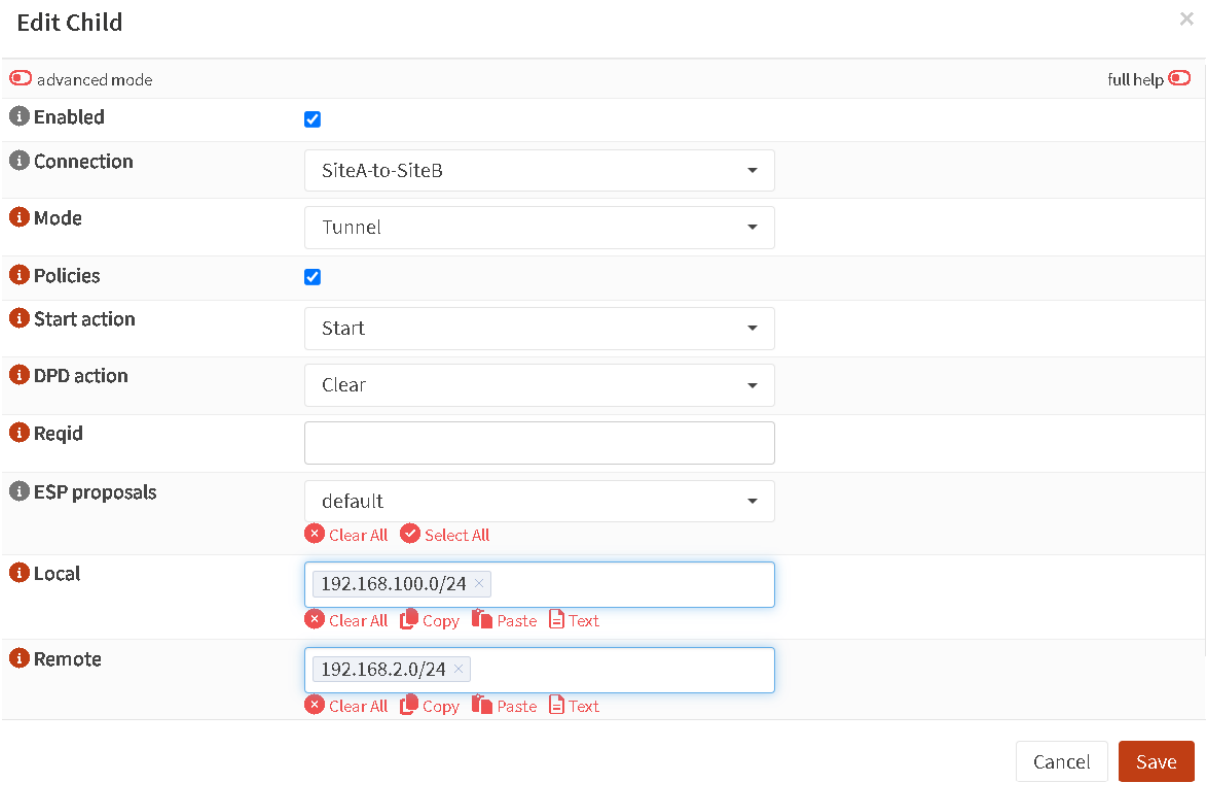
"Child SA defines which networks are protected by the IPsec tunnel. It tells each firewall which traffic should be encrypted and sent through the tunnel. This is also known as Phase 2 or the traffic selector."

What is Child SA?

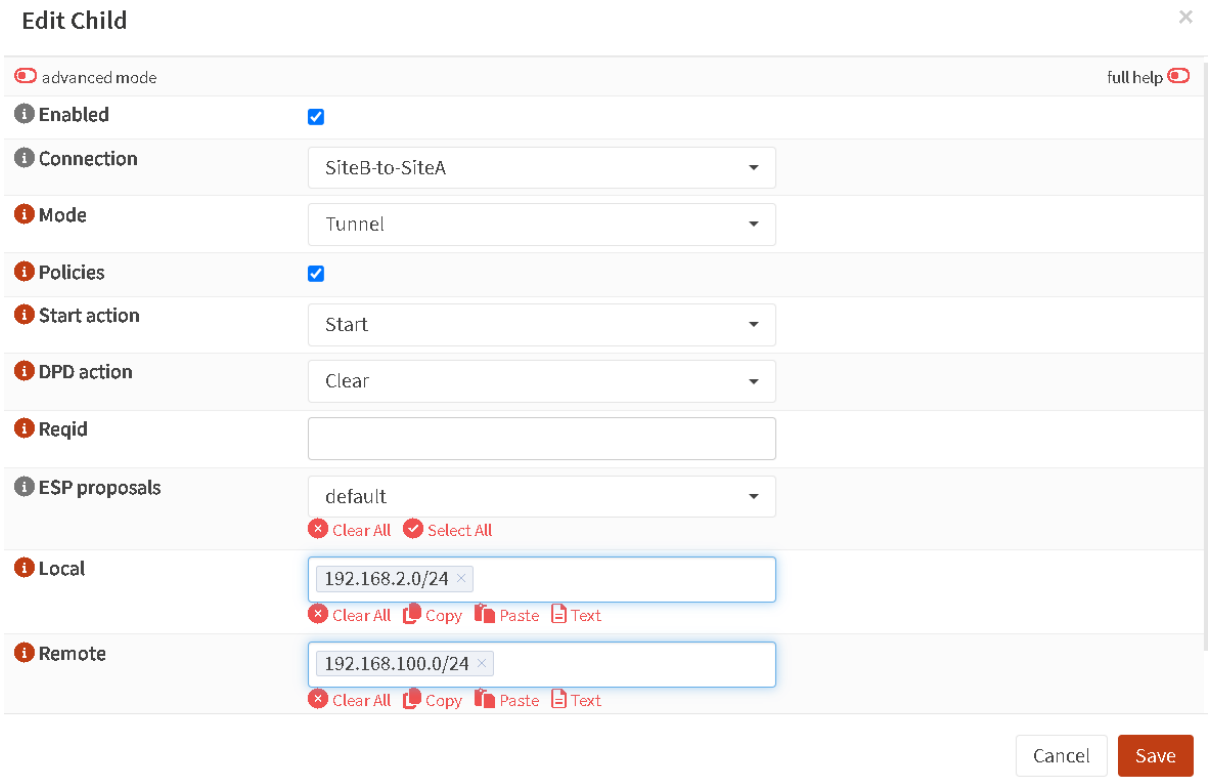
"While the IPsec Connection (Phase 1) establishes trust between two firewalls, the Child SA (Phase 2) defines the actual networks whose traffic will be encrypted. Think of it as: Phase 1 = two firewalls shake hands, Phase 2 = they agree on what data to protect."

Configuration on Site A:

- Inside **connection → scroll to Children → Add**



Configuration on Site B:



"Note: Local and Remote networks are swapped on Site B because each firewall refers to its own network as Local."

Why This Order Matters:

"IPsec tunnel configuration follows a logical sequence. Each step builds on the previous one — without completing all four steps, the tunnel cannot be established."

Step	Configuration	Question it Answers	Why it is Needed
1	Connection	Who do I connect to?	Defines the remote firewall's WAN IP and IKE version to use
2	Local Authentication	Who am I?	Tells the remote firewall your own identity so it can verify you
3	Remote Authentication	Who do I expect?	Tells your firewall which identity to accept from the remote side
4	Child SA	What traffic do I protect?	Defines which networks will be encrypted and sent through the tunnel

Enable IPsec & Apply

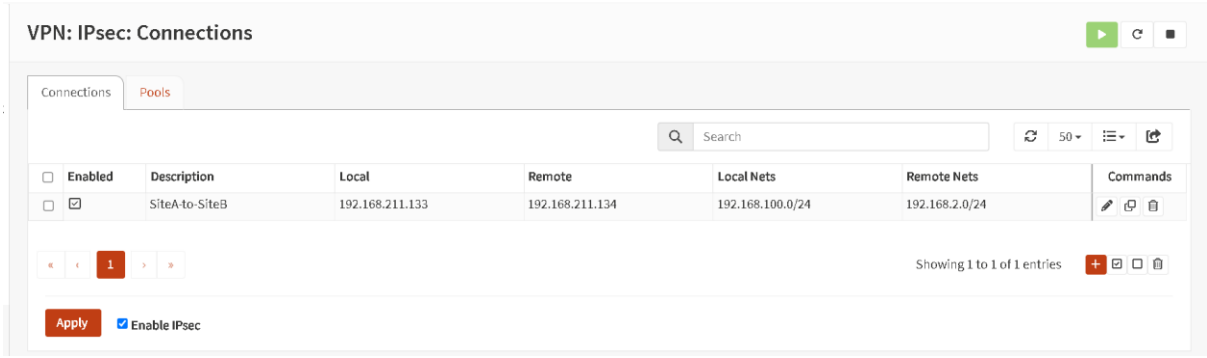
"After completing all configuration steps, IPsec service was enabled on both firewalls and changes were applied. This activates the tunnel and allows both firewalls to begin IKE negotiation."

Steps performed on both firewalls:

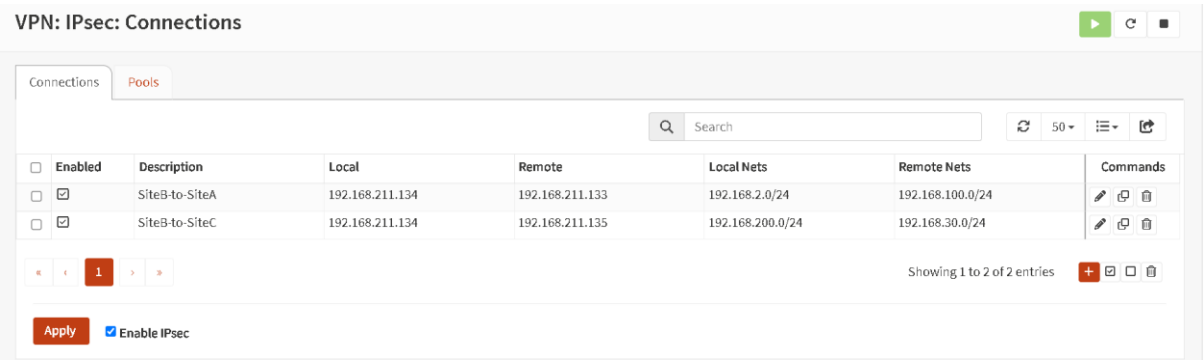
1. Go to **VPN → IPsec → Connections**
2. Check **"Enable IPsec"** checkbox
3. Click **"Apply"** button
4. OPNsense reloads IPsec service automatically

"Note: Both firewalls must have IPsec enabled and applied before the tunnel can be established. If only one side is enabled the tunnel negotiation will fail."

Site A:



Site B:



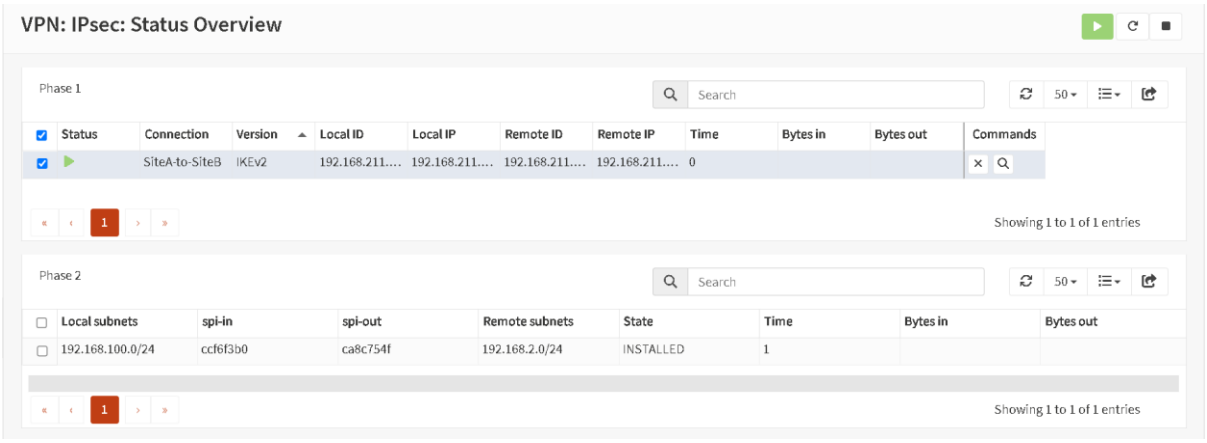
6.Verification & Testing

"After enabling IPsec on both firewalls, the tunnel status was verified using the OPNsense Status Overview page and ping tests were performed from Kali VM."

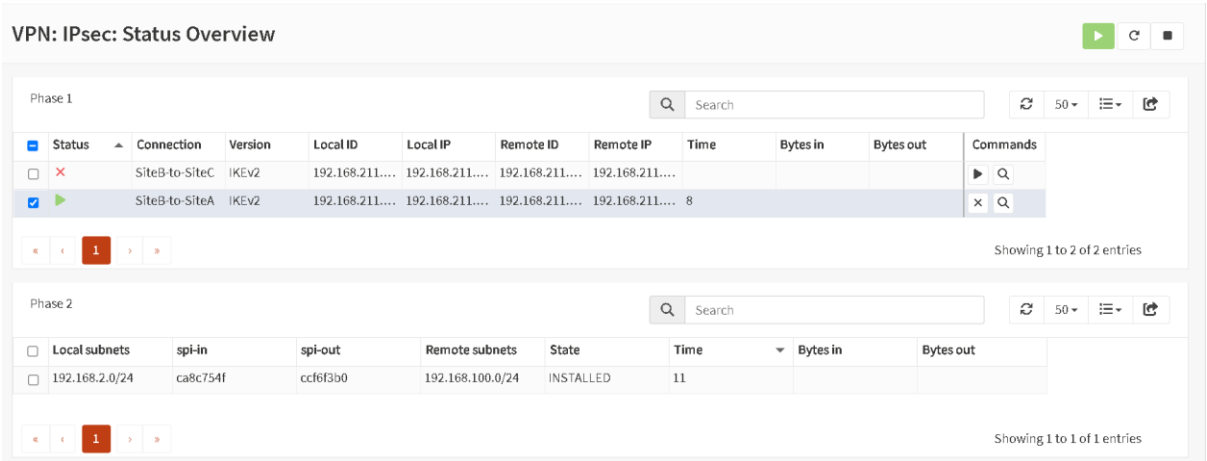
Tunnel Status Verification

"The IPsec Status Overview page shows the current state of Phase 1 and Phase 2 negotiations. A successful tunnel shows green arrow in Phase 1 and INSTALLED status in Phase 2."

Site A:

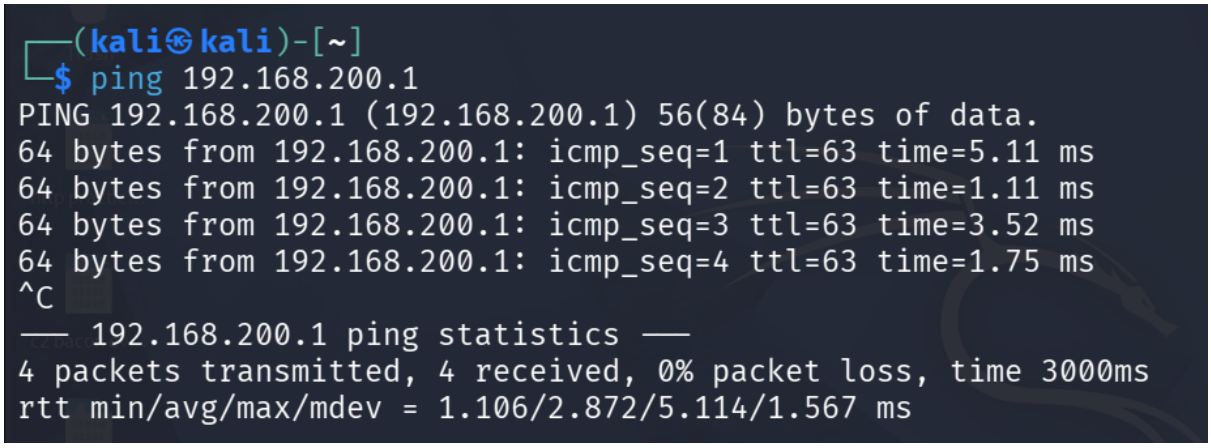


Site B:



Ping Test

"Ping test was performed from Kali VM (192.168.100.100) to Site B LAB_NET gateway (192.168.200.1) to verify traffic is passing through the tunnel successfully."



7. Challenges & Solutions

"During the configuration of the IPsec tunnel several challenges were encountered. Each challenge was investigated and resolved as documented below."

Challenge	Cause	Solution
Site B GUI not accessible via LAN	Windows host has no VMnet3 adapter — VMware Player free version limitation	Accessed Site B GUI via WAN IP (192.168.211.134) after running pfctl -d
GUI access lost after firewall reload	OPNsense WAN blocks GUI access by default after rule reload	Created Floating firewall rule on WAN interface allowing HTTPS (443) from 192.168.211.0/24
pfctl -d required every time before accessing Site B GUI	OPNsense processes automatic generated WAN rules before user defined rules. Default WAN deny rule blocks GUI access before our allow rule is checked. This is a known limitation when accessing OPNsense GUI through WAN interface instead of LAN	Floating rule was created as a partial fix but issue persisted due to VMware Player free version limitation. In production environments this issue does not occur as firewall GUI is always accessed through LAN or dedicated management interface — never through WAN

"Note: All challenges encountered were specific to the lab environment using VMware Player 17 free version. In a production environment these issues would not occur as firewalls are accessed through dedicated management interfaces."

8.Conclusion

"This report documents the successful configuration of a Site-to-Site IPsec VPN tunnel between two OPNsense firewalls using IKEv2 protocol with Pre-Shared Key authentication. The tunnel was verified using IPsec Status Overview showing Phase 1 established and Phase 2 INSTALLED on both firewalls. Ping tests from Kali VM confirmed that traffic is successfully passing through the encrypted tunnel."

Key Learnings:

Concept	What Was Learned
IKEv2	Modern protocol for tunnel negotiation between firewalls
Pre-Shared Key	Identical secret key on both sides for authentication
Phase 1	Firewalls authenticate each other and establish secure channel
Phase 2 / Child SA	Defines which networks are encrypted through tunnel
WAN IP in tunnel	Tunnel endpoints always use WAN IPs as they face the internet
Child SA swap	Local and Remote networks are swapped on each side

"This practical experience directly relates to real enterprise environments such as Tata Electronics where multiple country offices are connected via secure IPsec tunnels. The concepts of IKE negotiation, PSK authentication and traffic selectors are identical to those used in Palo Alto and Fortinet firewall configurations."